Cross Model Neighborhood Consistency

1. Goal

The objective of this week was to reproduce and audit baseline methods for measuring cross-model representational alignment. Specifically, we evaluate whether large language models (LLMs) exhibit consistent internal geometry across:

- Different architectures (Qwen3, OLMo3, Ministral3, Gemma3, LLaMA-32, Qwen2.5)
- Different datasets (WikiText vs COCO captions)
- Different layers (syntax-proxy vs semantic layers)
- Different subspaces (principal component projections)

Our goal is not to propose a new metric, but to ensure that baseline geometry measurements are:

1. Correctly implemented
2. Internally consistent
3. Robust across datasets
4. Interpretable

2. Baseline Metrics Reproduced

We reproduced and cross-validated the following alignment metrics:

2.1 Linear Representation Alignment

- **Centered Kernel Alignment (CKA)**
  Works across different embedding dimensionalities.
- **Orthogonal Procrustes Distance**
  Measures minimal rotation alignment (requires same dimensionality).

2.2 Neighborhood Geometry

- **Mutual k-NN Overlap**
  Measures shared neighborhood structure across models.

2.3 Cluster Agreement

- **Adjusted Rand Index (ARI)**
- **Normalized Mutual Information (NMI)**

2.4 Global Geometry Similarity

- **Distance Matrix Correlation (Spearman ρ)**
  Compares full pairwise distance structures.

2.5 Subspace Alignment

- PCA intrinsic dimensionality
- Per-component variance distribution
- Top-20 principal component subspace overlap

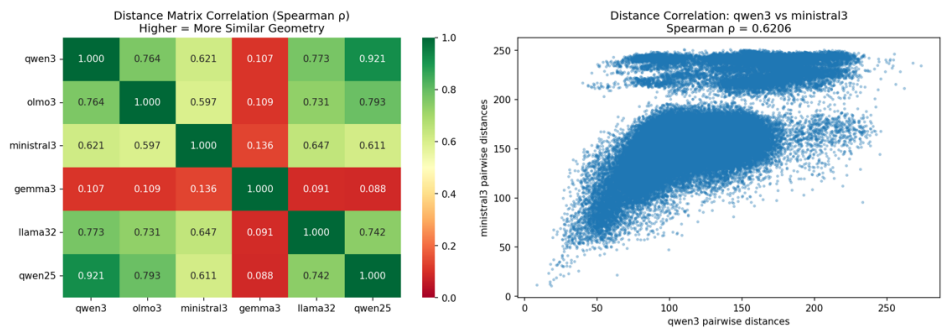
2.6 Syntax vs Semantics Subspace Comparison

Using Tenney-style layer convention:

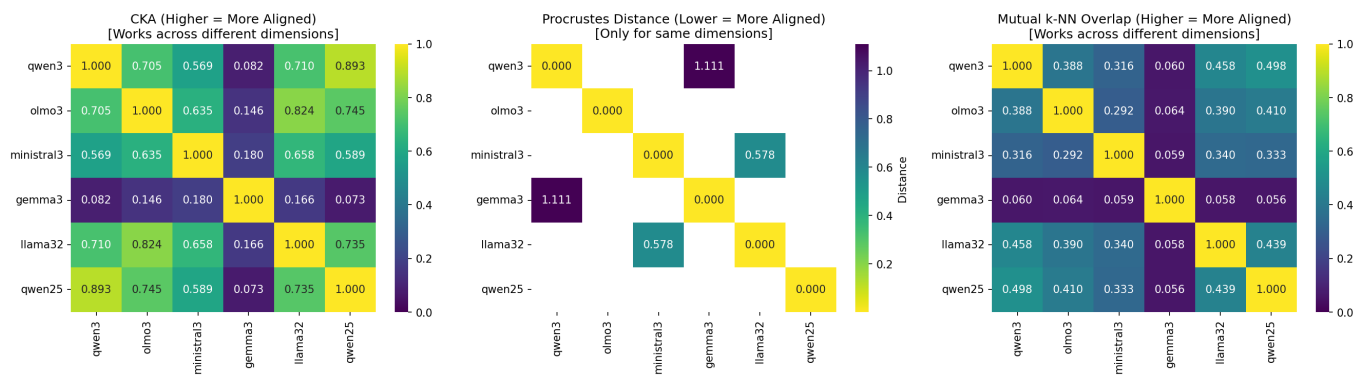
- Middle layer → syntax proxy
- Final layer → semantic proxy

3. Reproduction Results

3.1 Overall Alignment Structure



Across CKA, distance correlation, and neighborhood metrics:

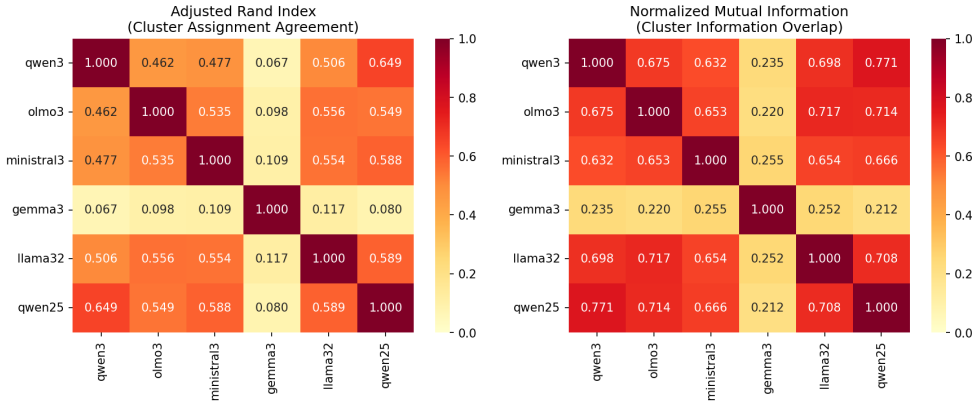


- **Qwen3, OLMo3, LLaMA-32, and Qwen2.5** exhibit strong geometric alignment.
- **Gemma3** consistently shows extremely low alignment across metrics.
- Ministral3 exhibits moderate alignment but lower structural consistency.

This pattern is stable across:

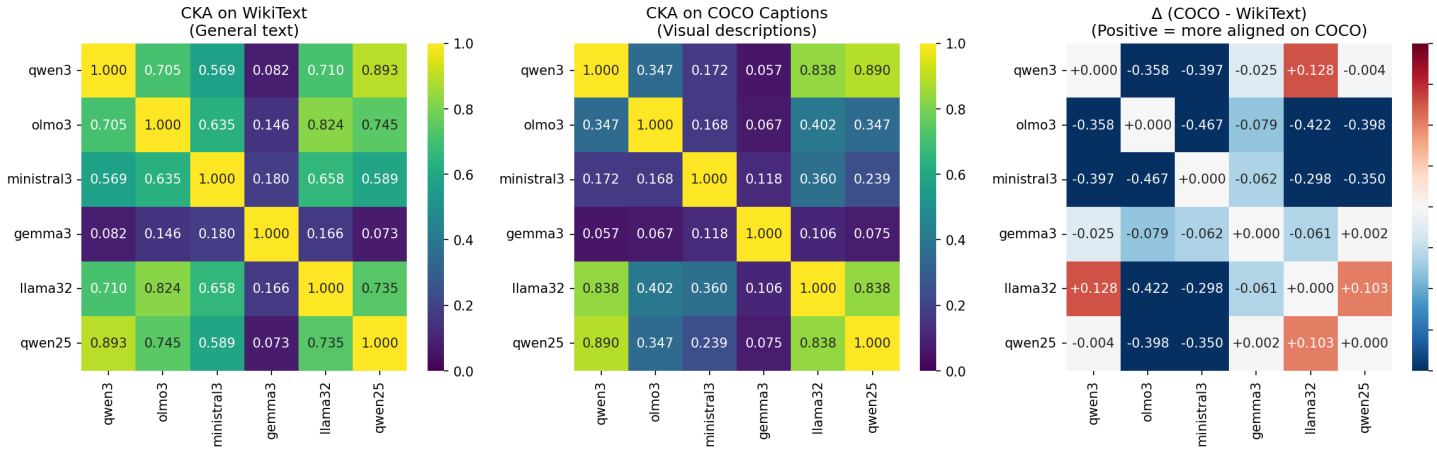
- CKA
- Distance correlation (ρ)

- Mutual k-NN overlap
- ARI / NMI clustering agreement



This confirms that cross-model geometry is not random: several models converge to similar internal structures.

3.2 Dataset Sensitivity (WikiText vs COCO)



When comparing alignment on:

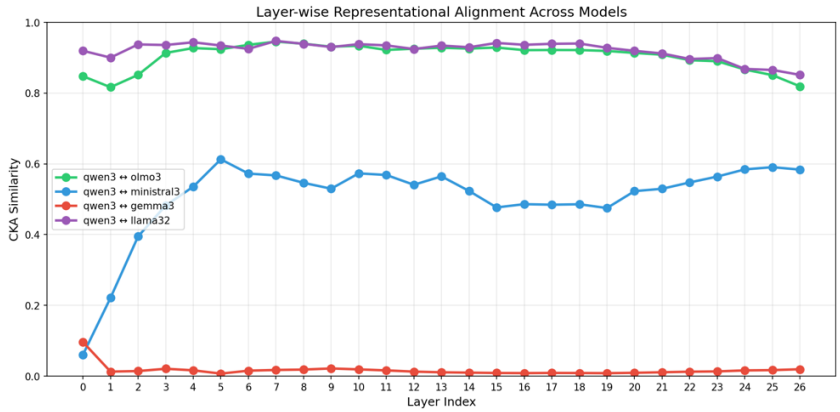
- WikiText (general text)
- COCO captions (visual descriptions)

We observe:

- Alignment among high-performing models slightly shifts under COCO.
- LLaMA-32 and Qwen3 increase alignment under visual caption data.
- OLMo3 and Ministral3 show decreased cross-model alignment under COCO.

This suggests representation geometry is dataset-dependent and partially modality-sensitive.

3.3 Layer-wise Behavior



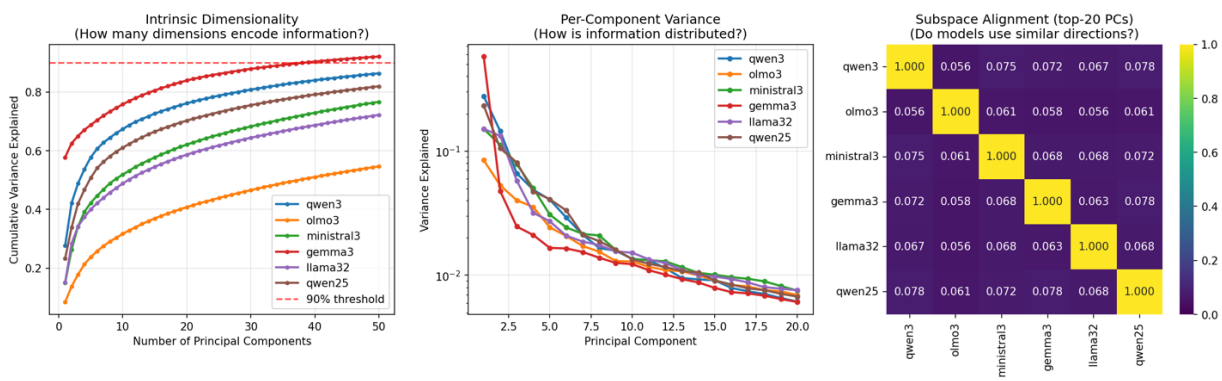
Layer-wise CKA reveals:

- Early layers rapidly converge to moderate alignment.
- Middle layers stabilize.
- Late layers show slight divergence.

This supports the hypothesis that:

- Lower layers encode more universal syntactic structures.
- Higher layers become more model-specific and semantically specialized.

3.4 Subspace Structure

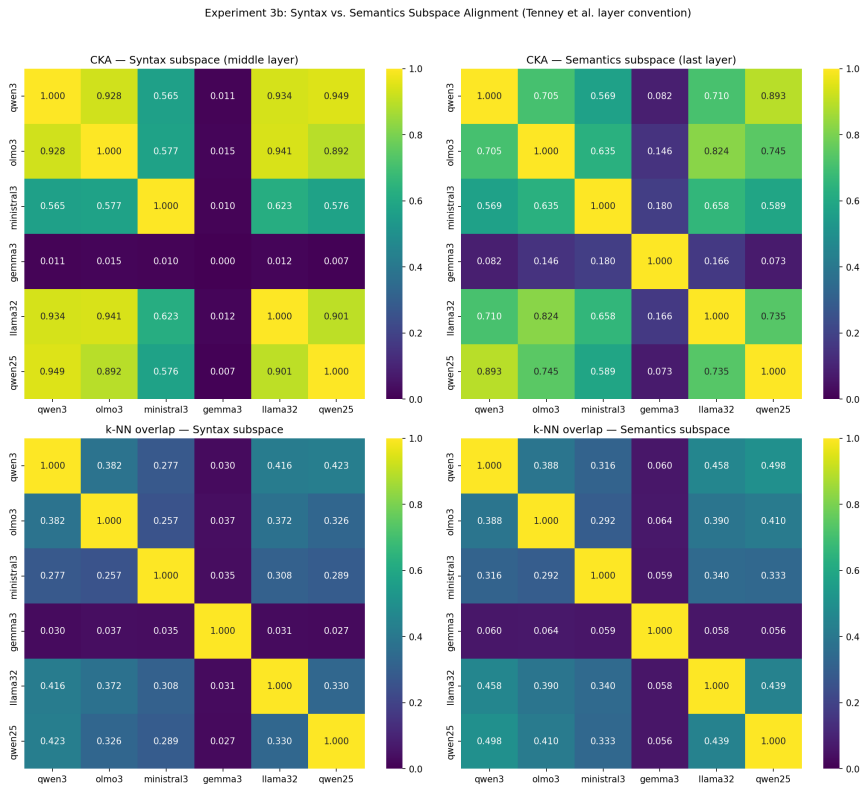


Intrinsic dimensionality analysis shows:

- Gemma3 concentrates variance heavily in first few PCs.
- Other models distribute variance more gradually.
- Top-20 PC subspace overlap is relatively low across all models.

This suggests that while global geometry aligns, principal directions differ.

3.5 Syntax vs Semantics Subspaces



When isolating:

- Syntax-proxy layer (middle)
- Semantic-proxy layer (final)

We observe:

- Strong cross-model alignment in syntax subspace among Qwen3, OLMo3, LLaMA-32, Qwen2.5.
- Very weak alignment for Gemma3 in syntax space.
- Moderate but more heterogeneous alignment in semantic space.

This indicates:

- Syntax geometry appears more universally structured.
- Semantic geometry is more model-specific.

4. Error Analysis

4.1 Gemma3 Representation Collapse

Across nearly all metrics:

- CKA ≈ 0 with other models

- Distance correlation extremely low
- ARI / NMI near random
- k-NN overlap minimal

Variance analysis shows:

- Heavy concentration in early principal components
- Rapid decay in per-component variance

This suggests:

- A representation geometry that is lower-rank or more collapsed
- Potential architectural or training differences affecting geometry

4.2 Dataset-Induced Drift

Under COCO captions:

- Some model pairs lose alignment
- Others gain alignment (notably LLaMA-32 with Qwen variants)

This suggests:

- Geometry may partially encode modality-specific structure
- Alignment is not fully architecture-invariant

4.3 Subspace Mismatch

Despite high CKA alignment in some model pairs:

- Top-20 PC subspace overlap remains low
- Principal axes differ even when global geometry correlates

This suggests:

- Models may encode similar relational structure
- But use different coordinate bases

5. Baseline Gaps

Despite strong baseline coverage, several limitations remain:

5.1 Linear Alignment Bias

CKA and Procrustes measure linear alignment.
They may fail to detect nonlinear isomorphic geometry.

Gap: Nonlinear manifold alignment not evaluated.

5.2 Clustering Dependence

ARI/NMI rely on:

- Choice of cluster count (k=10)
- Clustering algorithm (k-means)

Gap: Cluster structure may reflect algorithm bias rather than intrinsic geometry.

5.3 Dataset Scope

Only:

- WikiText (general text)
- COCO captions (visual descriptions)

Gap: No evaluation on:

- Dialogue
- Code
- Long-form reasoning
- Multilingual corpora

5.4 No Token-Level Behavioral Linking

We measure geometry but do not yet connect to:

- Hallucination
- Calibration
- Factual accuracy

Gap: Representational similarity is not yet tied to behavioral metrics.

5.5 Layer Selection Heuristic

Syntax vs semantics layer choice is heuristic.
True functional separation may not align cleanly with layer index.

Gap: No formal probe validation of layer semantics.

6. Conclusion

We successfully reproduced and audited a comprehensive suite of cross-model representational alignment baselines.

Findings:

- Strong geometric convergence exists across several major LLMs.
- Syntax geometry appears more universally shared.
- Semantic geometry is more model-specific.
- Dataset shifts induce measurable alignment drift.
- Some models (e.g., Gemma3) exhibit substantially different geometry.

The baselines are functioning, stable, and interpretable.

However, they remain limited by linear assumptions, clustering heuristics, and lack of behavioral grounding.

Future work will extend these baselines toward nonlinear alignment, task-conditioned geometry, and hallucination-aware representation analysis.